

CE 310 — Week 1 In-Class Exercise: The Flaw of Averages

Tucson International Airport, NOAA Hourly Weather Data — 2023 (8,760 Hours)

Before You Begin

Download both files from D2L into the **same folder** on your computer:

- Week1_Tucson_Weather_2023.csv — the raw dataset (8,760 rows)
- Wk01.04_In_Class_Answer_Sheet.xlsx — your submission template

Answer Sheet layout — six columns, most pre-filled:

Col	Role	You fill this?
A	Question label and its point breakdown	No — pre-filled
B	Question description, and a hint for the formula	No — pre-filled
C	Your Excel formula — the formula itself, not a description of it	Yes — 1 pt in Section B and on A8; read but not scored elsewhere in Section A
D	Intermediate value (★ questions only)	Yes, if ★
E	Your numerical answer	Yes
F	Unit	No — pre-filled

Example — how a completed row looks. The two rows below are a format illustration only. They are not questions on your answer sheet, and the numbers are invented — they come from a different building's data, not the Tucson file.

Col A	Col B	Col C	Col D	Col E	Col F
X1	Average daily water use	'=AVERAGE(B2:B366)		41.3	gal/day
X2	Month with the highest average use	'=AVERAGEIF(. . . 63.8		8	month #

Type your formula as text. In Col C, put an apostrophe before the = so Excel stores the formula instead of running it — for example, '=AVERAGE(B2:B8761). Without the apostrophe Excel evaluates it and your method is lost. **Screenshots are not accepted.**

Col C is read, not just checked for something. It is matched against the function the question asks for and against the values that make it *this* question rather than the one beside it: '=COUNTIF(B2:B8761, ">95") answers A6, and the same formula with ">100" does not. A description of what you did — “computed in Excel”, “see attached” — is not a formula and earns nothing. **The formula is worth 1 point on every Section B question, whether the question is worth 4 or 5, and you lose it even when your number in Col E is right.**

Excel Quick Reference

What you want	Function
Mean	=AVERAGE(range)
Median	=MEDIAN(range)
Count rows in a band (two conditions)	=COUNTIFS(range, ">=80", range, "<90")
Maximum	=MAX(range)
Minimum	=MIN(range)
Percentile — k=1 for Q1, k=3 for Q3	=QUARTILE(range, k)
Sample standard deviation	=STDEV(range)
Count rows matching one condition	=COUNTIF(range, ">95")
Count rows matching two conditions	=COUNTIFS(col1, ">80", col2, ">50")
Mean for rows matching a condition	=AVERAGEIF(crit_col, value, avg_col)
Nth largest value	=LARGE(range, n)
Month number from a date	=MONTH(date_cell)
Cond. std dev — all Excel (Ctrl+Shift+Enter)	=STDEV(IF(month_col=7, data_col))
Cond. std dev — Excel 365	=STDEV(FILTER(data_col, month_col=7))

Setup — Import the Data (~10 min)

You do all of this week's work in **one file: your Answer Sheet**. The data, your scratch work, and your charts all live there, so everything you built is in the file you hand in.

1. Open your Answer Sheet and add a new sheet. Name it Data. (Right-click any sheet tab -> **Insert** -> **Worksheet**.) Add as many sheets as you like — only Instructions, Answers, Work, Charts and Written have to keep their names.
2. Click cell **A1** of the Data sheet, then go to **Data -> Get Data -> From Text/CSV**, select **Week1_Tucson_Weather_2023.csv**, and click **Load To... -> Existing worksheet**. Start at A1 — every formula in this handout assumes it, so if your data begins somewhere else your column letters will not match the examples.
3. Confirm you have **8,760 data rows** (plus one header row).
4. Columns: DateTime, Temp_F, Wind_Speed_mph, Relative_Humidity_pct.
5. In cell **E1** — the first empty column after the four imported ones — add the header Month. In E2, enter =MONTH(A2), then fill down to row 8,761. Data column B is now Temp_F and column E is Month, which is what every formula below refers to.

Your file will grow to a megabyte or two once the data is in it. That is expected — upload it as-is.

SECTION A — Fundamentals (~25 min)

Section A builds the toolkit one function at a time — a mean, a maximum, a spread, a count. The goal is knowing what each statistic measures before you ask it to carry a design decision.

Round temperatures to 2 decimal places. Round counts to whole numbers.

A1 (1 pt) — Annual Average Temperature

Compute the mean of all 8,760 hourly temperature readings.

A2 (1 pt) — Annual Maximum Temperature

Compute the single highest temperature recorded in 2023.

A3 (1 pt) — Annual Minimum Temperature

Compute the single lowest temperature recorded in 2023.

A4 (1 pt) — Annual Temperature Range

Compute MAX – MIN across all 8,760 hours.

A range of ~84°F against an average of ~72°F — the dataset spans from near-freezing to desert-summer extremes.

A5 (1 pt) — Annual Standard Deviation of Temperature

Compute the **sample** standard deviation of all hourly temperature readings.

A standard deviation of ~18°F against a mean of ~72°F is the first warning that “design for the average” will fail.

CV has no Answer Sheet cell and no points of its own. Once you have A1 and A5, work out CV = $(A5 \div A1) \times 100\%$ in Work!B195, the *CV check* block at the foot of the Work sheet. Paragraph 1 of the memo asks for it, and that is where it earns credit.

A6 (1 pt) — Hours Above 95°F

Count the hours where temperature exceeded 95°F.

A7 (1 pt) — Hours Above 100°F

Count the hours where temperature exceeded 100°F.

A8 ★ (4 pts) — Month with Highest Average Temperature

Find which calendar month (1–12) had the highest average temperature.

You need all twelve monthly averages — A8 asks for the hottest one, and Chart 1 plots the whole set.

Where they go. Open the Work sheet. The first block is headed **Question A8 + Chart 1 — Monthly average temperature**, with the month names already filled in; type your twelve averages into the yellow cells beside them, B7 down to B18. Do not build this table anywhere else — Chart 1 is graded by clicking the chart to see which cells it plots, and these are the cells it should point at.

What to type. One AVERAGEIF per row, reading your Data sheet. In B7, for January:

=AVERAGEIF(Data!\$E\$2:\$E\$8761, 1, Data!\$B\$2:\$B\$8761)

Then copy that formula down to B18, changing the month number on each row — 1 for January through 12 for December. The month number is the only thing that changes.

Three things that trip people up here:

- **Data! is not optional.** The table lives on Work and the data lives on Data, so every range needs the sheet name in front of it. Without it Excel looks on Work, finds nothing, and returns #DIV/0!.
- **Lock the ranges with \$** so they don't shift as you fill down. The month number is the only thing that should change from row to row.

- **\$E\$ is the Month column, \$B\$ is Temp_F.** That is what Setup step 5 built, so the formula reads: look down the month numbers, match this month, average the matching temperatures. If your import did not start at A1 of Data, fix that first rather than rewriting the formula — everything else in this handout assumes it too.

Once the twelve values are in, read off the largest and which month it belongs to. Then scroll back up to the A8 row and record it:

- **Col C:** One of the AVERAGEIF formulas from your A8 table — not a formula for the month number itself
- **Col D:** Average temperature of the hottest month
- **Col E:** Month number (1–12)

A9 (1 pt) — Annual Median Temperature

Compute the median of all 8,760 hourly temperatures.

Compare to A1. Similar values suggest a symmetric distribution; a notable gap suggests skew.

A10 (1 pt) — Q1: 25th Percentile Temperature

Compute the first quartile of all hourly temperatures.

A11 (1 pt) — Q3: 75th Percentile Temperature

Compute the third quartile of all hourly temperatures.

A12 (1 pt) — Interquartile Range (IQR)

Compute $Q3 - Q1$.

A13 (1 pt) — Modal Bin (Most Common 10°F Band)

Count the hours whose temperature falls in the **80–90°F** band: at or above 80, below 90.

Why a band and not a single value. MODE on 8,760 continuous readings returns 81°F — a value that occurs 180 times, about 2% of the year. That is a rounding artefact, not the shape of Tucson's climate. For continuous data the meaningful question is which *interval* is most common, which is exactly what a histogram bar is. You will build that histogram in Week 2; this is its tallest bar.

SECTION B — In-Depth Analysis

Section B puts the toolkit to work. Every question here ends in a design number — the temperature to size for, the wind pressure to resist, the moisture to remove — and the statistic is only how you get there.

B1 (4 pts) — ASHRAE 1% Design Dry-Bulb Temperature

The cooling design temperature is defined as the **88th-highest** hourly temperature of the year (1% of 8,760 hours = 87.6, rounded to 88).

Use the Function Reference to find the appropriate function.

B2 (4 pts) — Design Temperature Minus Annual Average

How much hotter is the ASHRAE design temperature than the annual average? This is how many degrees an average-based HVAC system would be undersized.

B3 ★ (5 pts) — Standard Deviation of Temperature in July

How much does temperature vary hour-to-hour within July? STDEV alone cannot filter by month — use a conditional std dev from the Function Reference.

Two options — try Excel 365 first; if it returns #NAME?, use the all-Excel version:

- **Excel 365:** =STDEV(FILTER(B2:B8761, E2:E8761=7)) — press Enter normally
- **All Excel:** =STDEV(IF(E2:E8761=7, B2:B8761)) — press **Ctrl+Shift+Enter** (not just Enter)

When the array formula works correctly, curly braces appear automatically around it:

Before: =STDEV(IF(E2:E8761=7, B2:B8761))

After: {=STDEV(IF(E2:E8761=7, B2:B8761))}

If you typed the curly braces yourself, the formula will not work — they must appear on their own after Ctrl+Shift+Enter.

- **Col D:** July average temperature — confirms you filtered the right month
- **Col E:** July standard deviation (°F)

B4 ★ (5 pts) — Standard Deviation of Temperature in January

Same approach as B3, for January (month = 1).

- **Col D:** January average temperature
- **Col E:** January standard deviation (°F)

Which month is more variable? Think about what drives that variability.

B5 (4 pts) — Average Temperature in July

Compute the average temperature for July hours only.

B6 (4 pts) — Average Temperature in January

Compute the average temperature for January hours only.

Questions B7–B11 — Wind Loads (ASCE 7-22)

The same flaw appears in structural engineering. The formula $P = 0.00256 \times C_a \times V^2$ governs wind pressure on every building and sign structure. Designing for average wind speed instead of peak produces dangerously undersized structures.

B7 (4 pts) — Annual Average Wind Speed

Compute the mean of all 8,760 wind speed readings (Column C).

B8 (4 pts) — Annual Maximum Wind Speed

Compute the single highest wind speed recorded in 2023.

B9 ★ (5 pts) — Wind Pressure at Annual Average Speed

$P = 0.00256 \times C_a \times V^2$ where $C_a = 1.3$, V in mph, P in psf.

Apply this to your average speed from B7.

- **Col D:** V^2 (your B7 value, squared)

- **Col E:** P in psf, rounded to 3 decimal places

B10 ★ (5 pts) — Wind Pressure at Annual Maximum Speed

Same formula, using your maximum speed from B8.

- **Col D:** V^2 (your B8 value, squared)
- **Col E:** P in psf, rounded to 3 decimal places

B11 (4 pts) — Ratio of Peak to Average Wind Pressure

How many times larger is the peak-gust wind pressure than the pressure at average wind speed?

- **Col E:** The ratio, rounded to 2 decimal places

Col D stays blank here. It used to ask for your B10 value — the number B10 already scored — so the ratio is a single 4-point question now, like the rest of Section B.

This ratio is how many times the design load a billboard sized for average wind would experience during the peak gust.

Questions B12–B14 — Latent Load (ASHRAE 62.1)

HVAC systems must remove moisture from outdoor air. An engineer who sizes dehumidification for the annual average humidity instead of peak monsoon conditions will produce a system that fails when it is needed most.

B12 (4 pts) — Annual Average Relative Humidity

Compute the mean of all 8,760 relative humidity readings (Column D).

B13 (4 pts) — Hours Where Relative Humidity Exceeds 60%

Count the hours where outdoor RH exceeded 60%.

Tucson's North American Monsoon pushes RH above 60% for over 1,000 hours per year — a desert city with a real dehumidification load.

B14 (4 pts) — Hours with Warm Temperatures AND High Humidity

Count the hours where temperature exceeded 80°F **and** relative humidity exceeded 50% simultaneously. Refer to the Function Reference for the two-condition count function.

Required Charts (15 pts)

Both charts belong on the **Charts sheet** of your Answer Sheet, in the outlined boxes. That sheet lists what each chart is graded on.

Build each chart from its own table on the Work sheet — Chart 1 from the A8 monthly table, Chart 2 from the 168 July rows. Then right-click the chart -> **Move Chart** -> **Object in** -> Charts, and drag it into the matching box.

Leave both as live charts — do not replace either with a picture of itself. Your preceptor clicks the chart to see which cells it plots, which is how a correct chart built from the wrong column gets caught.

Chart 1 (7 pts) — Monthly Average Temperature

Select the A8 table you filled in on the Work sheet — A6:B18, headers included — and insert a line or column chart. Months 1–12 on the X-axis, monthly average temperature on the Y-axis. Add a horizontal reference line at the annual average (A1). Title: Monthly Average Temperature – Tucson, AZ 2023.

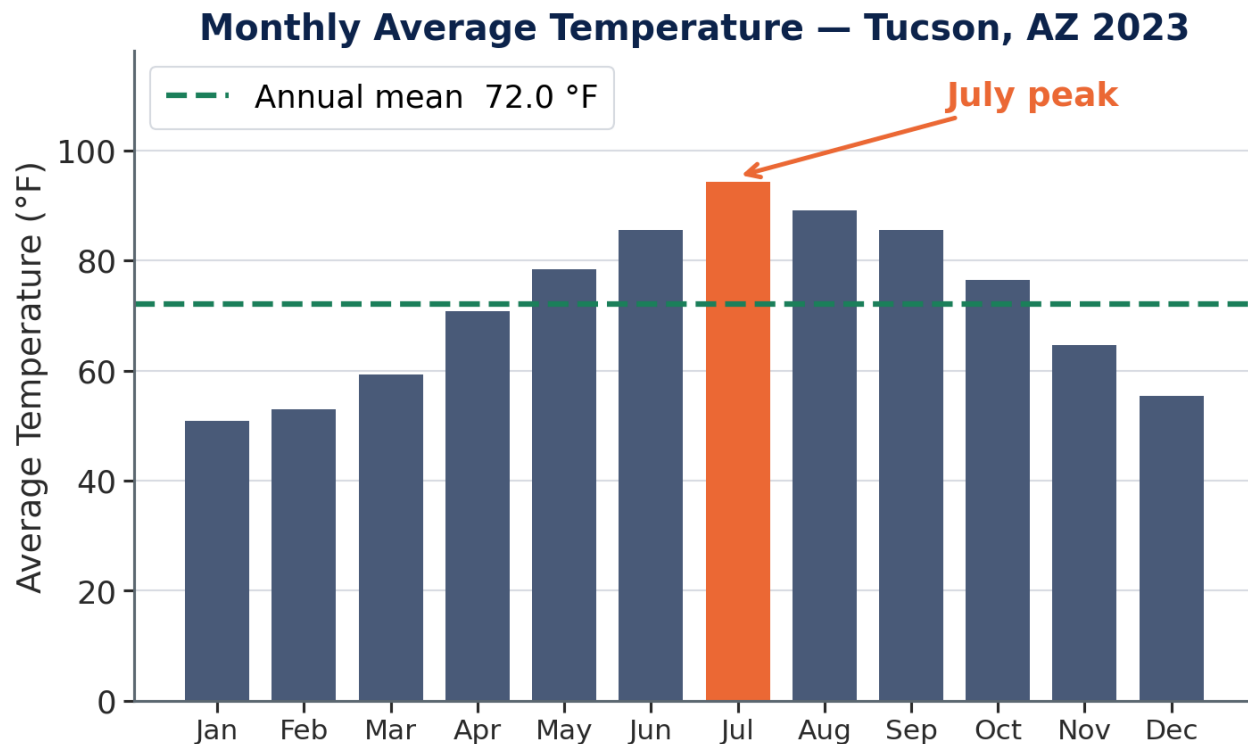


Figure 1: Target shape for Chart 1

Chart 2 (8 pts) — Hourly Temperature vs. Annual Average: July 1–7

Step 1 — Extract 168 rows for July 1–7:

1. Click any cell in your data table -> **Data -> Filter**.
2. Click the dropdown on DateTime -> **Date Filters -> Between**.
3. Enter 7/1/2023 and 7/7/2023 -> **OK**. You should see 168 rows.
4. Copy the filtered Temp_F column -> paste (Values only) into Work!B23, in the block headed *Chart 2 — July 1–7 hourly temperature*. The 168 hour numbers are already down column A; your temperatures fill column B.
5. Turn the filter off: **Data -> Filter** again.

(If date filters don't work in your Excel version, ask your preceptor for the helper-column method.)

Step 2 — Build the chart:

1. Fill Work!C23:C190 — the Annual avg (°F) column beside your pasted temperatures — with your A1 answer, the same value all 168 rows.
2. Select B22:C190 -> **Insert -> Line Chart**.
3. Format Annual_Avg as a dashed red line.
4. Title: Hourly Temperature vs. Annual Average – Tucson, July 1–7, 2023. Label both axes.

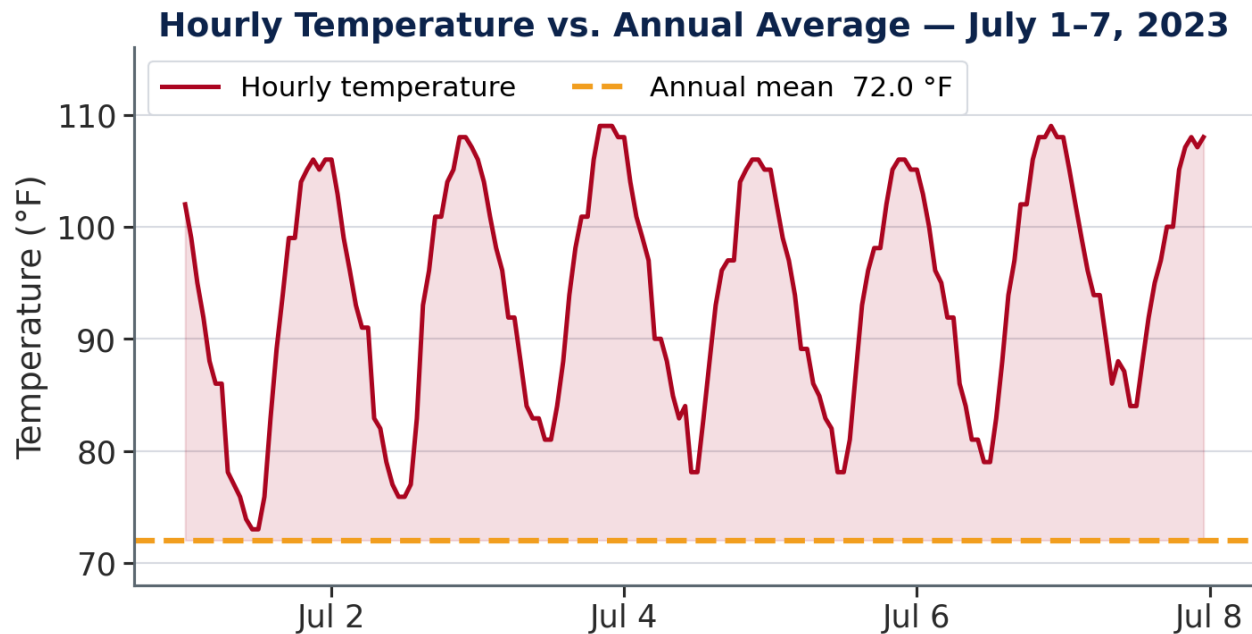


Figure 2: Target shape for Chart 2

Memo Deliverable (9 pts)

Type your memo in the **Written sheet** of the Answer Sheet (yellow cells P1, P2, P3).

To: Building Systems Design Client **From:** [Your Name] **Subject:** Why Annual Average Temperature Is Insufficient for HVAC Design — Tucson, AZ

Paragraph 1 (3 pts) — The averages problem. State the annual average (A1), annual maximum (A2), and CV ($= A5 \div A1 \times 100\%$). Use the CV value and the three-tier threshold from lecture ($< 10\%$ reliable / $10\text{--}20\%$ report σ / $> 20\%$ use percentiles) to explain whether the mean alone is a reliable design value for Tucson's temperature. Reference B2 for the gap.

Paragraph 2 (3 pts) — The design temperature. State the ASHRAE design temperature (B1) and hours above 100°F (A7). Include Chart 2.

Paragraph 3 (3 pts) — Humidity burden. Using B12 and B13, explain why annual-average humidity understates the dehumidification load in Tucson.

Submission: Save as CE310_W1_<NetID>.xlsx and upload to **D2L → Assignments**. Enter your NetID in the **NetID** cell at the top of the Answers sheet — that cell is what matches your work to your student record. A 5-point deduction applies if it is blank.

Before You Submit

- ☐ Full name and NetID entered on the Answer Sheet — a blank NetID cell takes a 5-point deduction
- ☐ File saved as .xlsx and named CE310_W1_<NetID>.xlsx
- ☐ Sheet tabs not renamed, reordered, or deleted
- ☐ Every answer typed into its designated cell in **Col E** — answers placed anywhere else receive no credit
- ☐ Your formula typed as text in **Col C**, with an apostrophe before the = — the formula itself, not a description of it. It is 1 point on every Section B question, and on A8. Elsewhere in Section A it is not scored, but write it anyway: your preceptor reads it, and a blank one beside a correct answer gets flagged
- ☐ ★ questions also have their intermediate value in **Col D**

- ☐ Both charts sitting in their boxes on the Charts sheet, as live charts and not pictures, with titles and axis labels
 - ☐ Memo written in the Written sheet
 - ☐ Uploaded to **D2L → Assignments** by **9:00 AM, Wed Sep 2**
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Grading Summary

Every question is worth a whole number of points.

- **Section A** — 1 pt each, on the answer. The exception is **A8** at 4 (1 formula + 1 intermediate + 2 answer), the one Section A question that needs a helper table.
- **Section B** — 4 pts each (1 formula + 3 answer), or **5** for a ★ (1 formula + 1 intermediate + 3 answer).

Component	Points
Section A: A1–A13	16
Section B: B1–B14	60
Chart 1 — Monthly averages	7
Chart 2 — July hourly vs. annual average	8
Memo — 3 paragraphs referencing specific values	9
Total	100